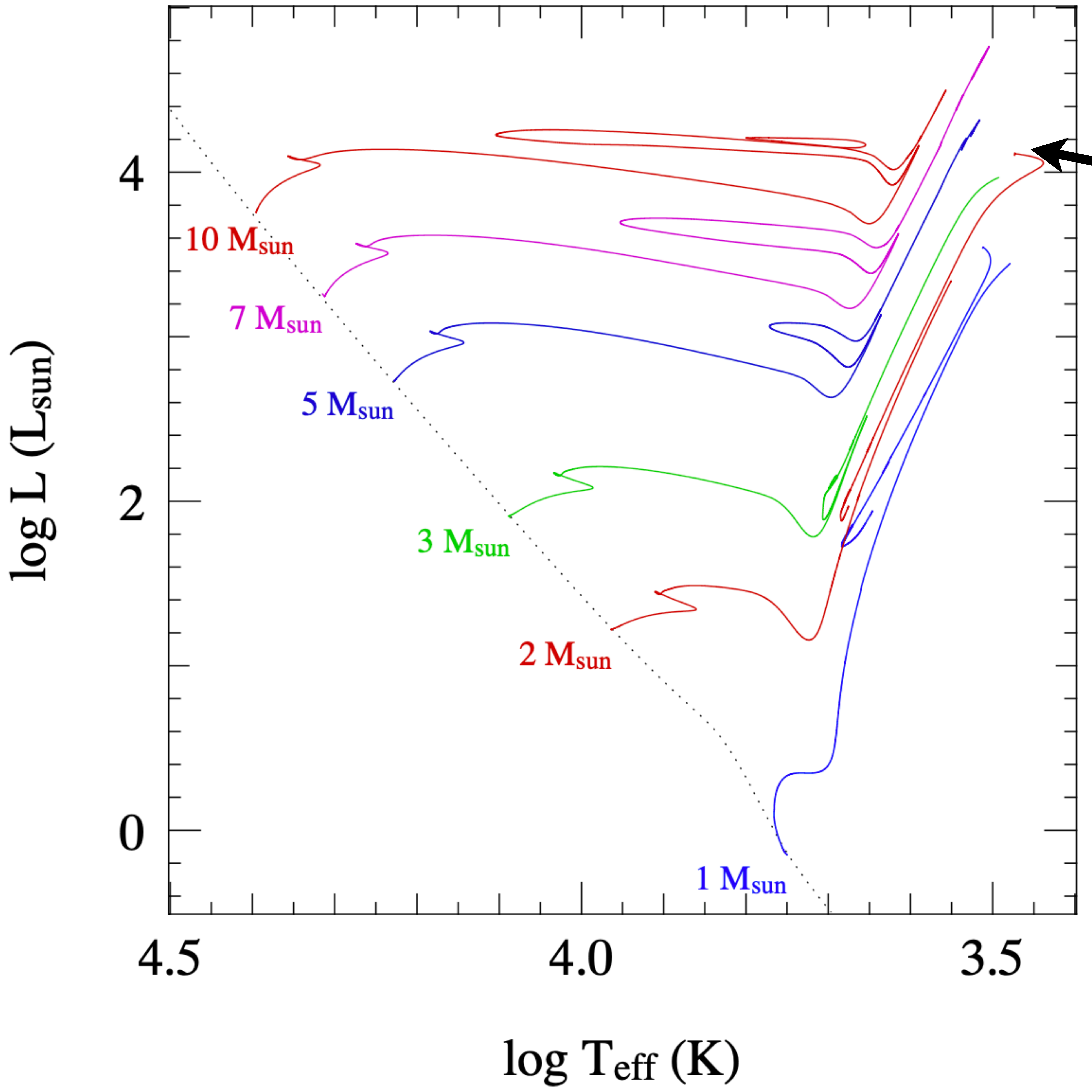
Stellar remnants

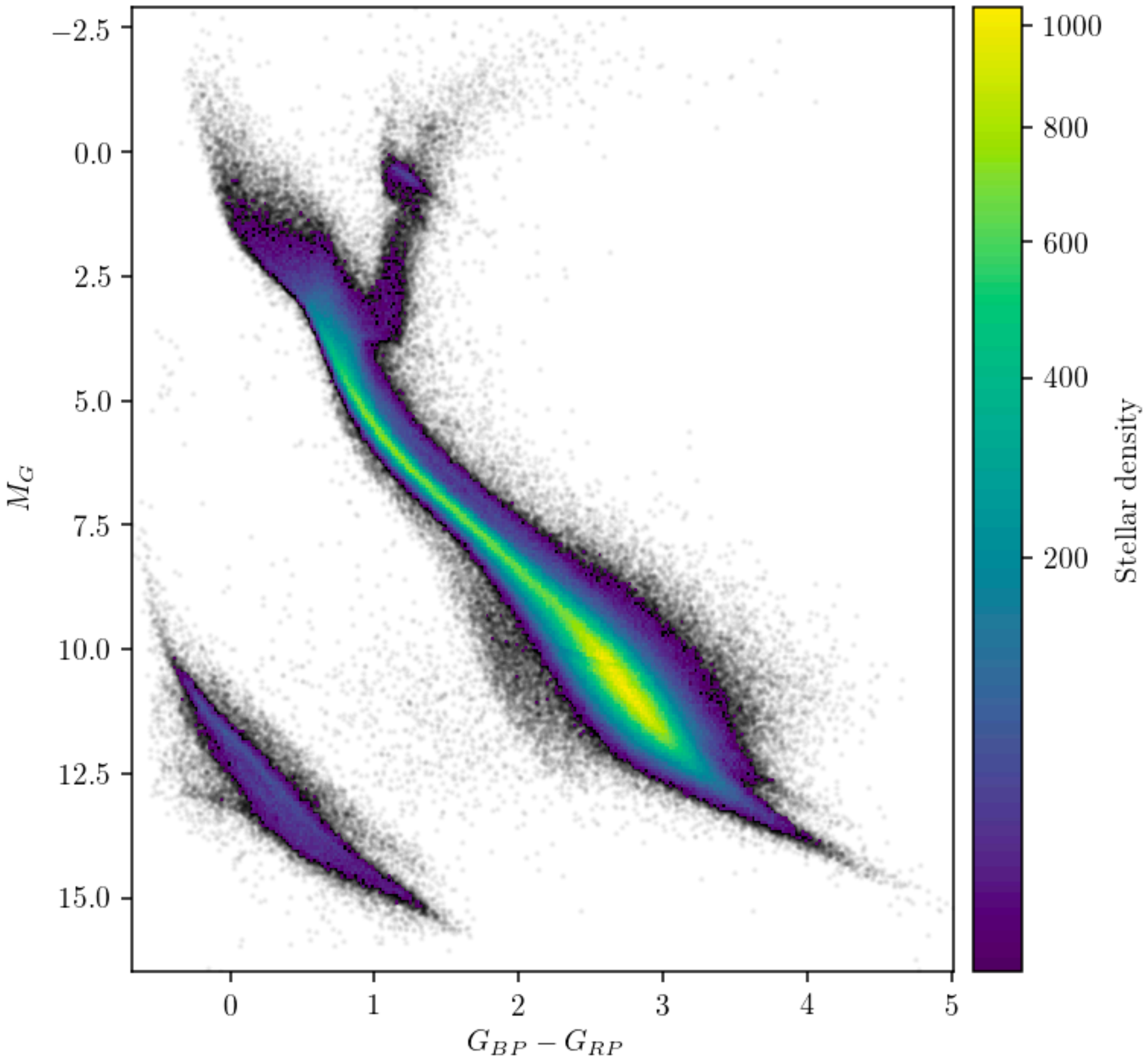
But first… some announcements!

1) projects are up on the projects page on canvas —
you should also have seen an announcement about this!
    1a) new project selection must be made before class next Tuesday on Piazza
    1b) you must work with a new partner
2) Mid-semester grades are in — y'all are killing it!
3) HW 5 will be posted tonight — stellar evolution & remnants
4) If you want to learn about gravitational waves & get a sneak peek for a future lecture (or eat some free pizza) you can come to Doherty Hall A302 today at 5pm to listen to me again.



During the AGB, superwinds lead to stripping of the hydrogen envelope

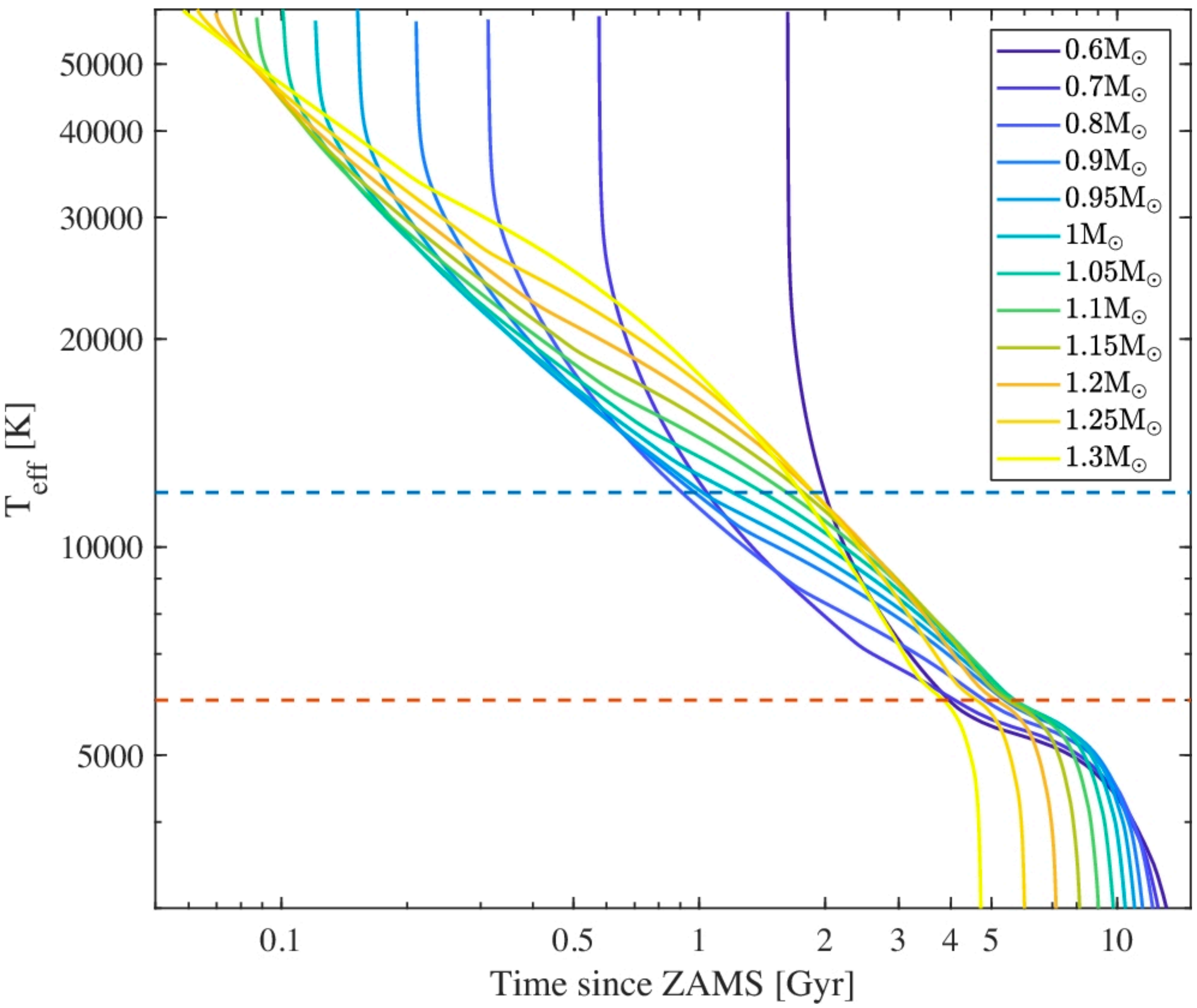
This causes the star to move toward hotter temps and lower luminosities



Eventually stars end up in the lower left part of the HR Diagram (or CMD as in the left figure) and remain in that region as white dwarfs ~forever~

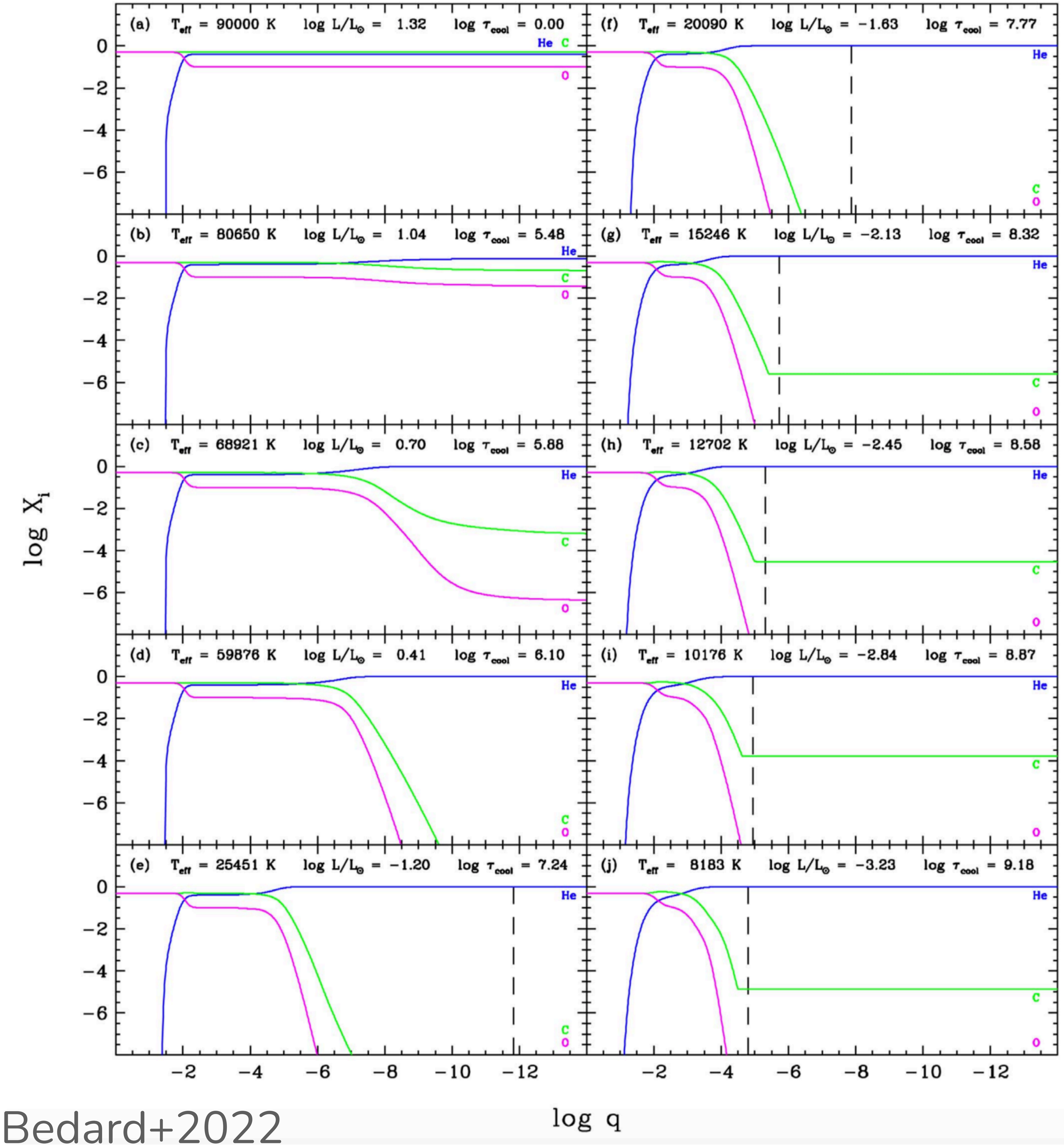
Low mass white dwarfs are brighter because they are larger than more massive white dwarfs

Maoz, Hallakoun, Badenes 2018



White dwarfs do not undergo any burning (at least as single WDs) so they only cool over time

WDs cool with mass-dependent rates since more massive WDs originate from more massive stellar cores



Bedard+2022

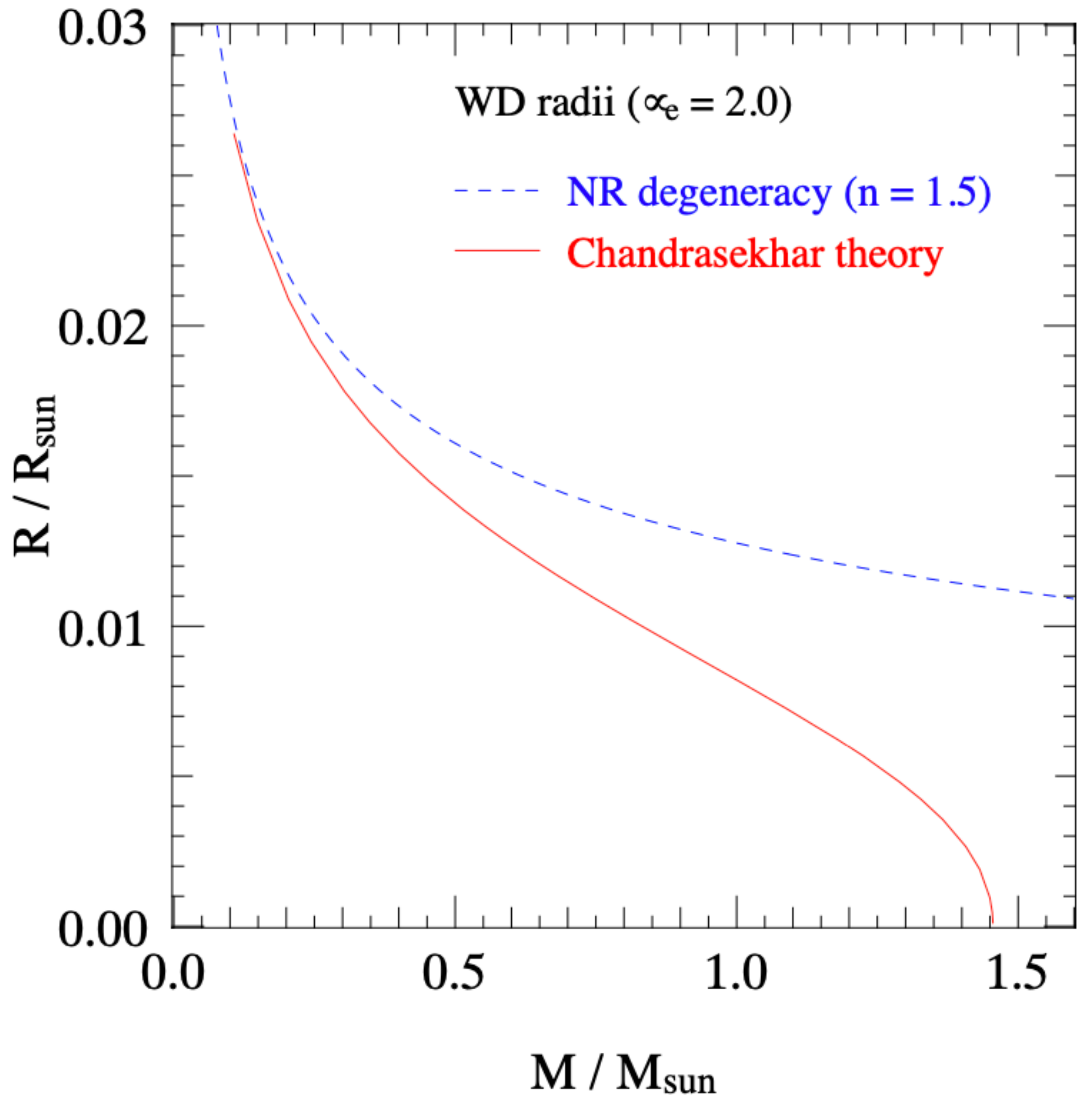
Elements in the core undergo stratification due to gravitational settling

Dotted line shows the base of the convective zone; q = 1-m/M is the fractional mass depth

WD structure: supported by electron degeneracy pressure

$$P_e = \frac{1}{3}\int_0^{p_F} \frac{8\pi p^4}{h^3 m_e}\,dp = \frac{8\pi}{15h^3 m_e}\,{p_F}^5 = \frac{h^2}{20m_e}\left(\frac{3}{\pi}\right)^{2/3} {n_e}^{5/3}$$

<— no temperature dependence!



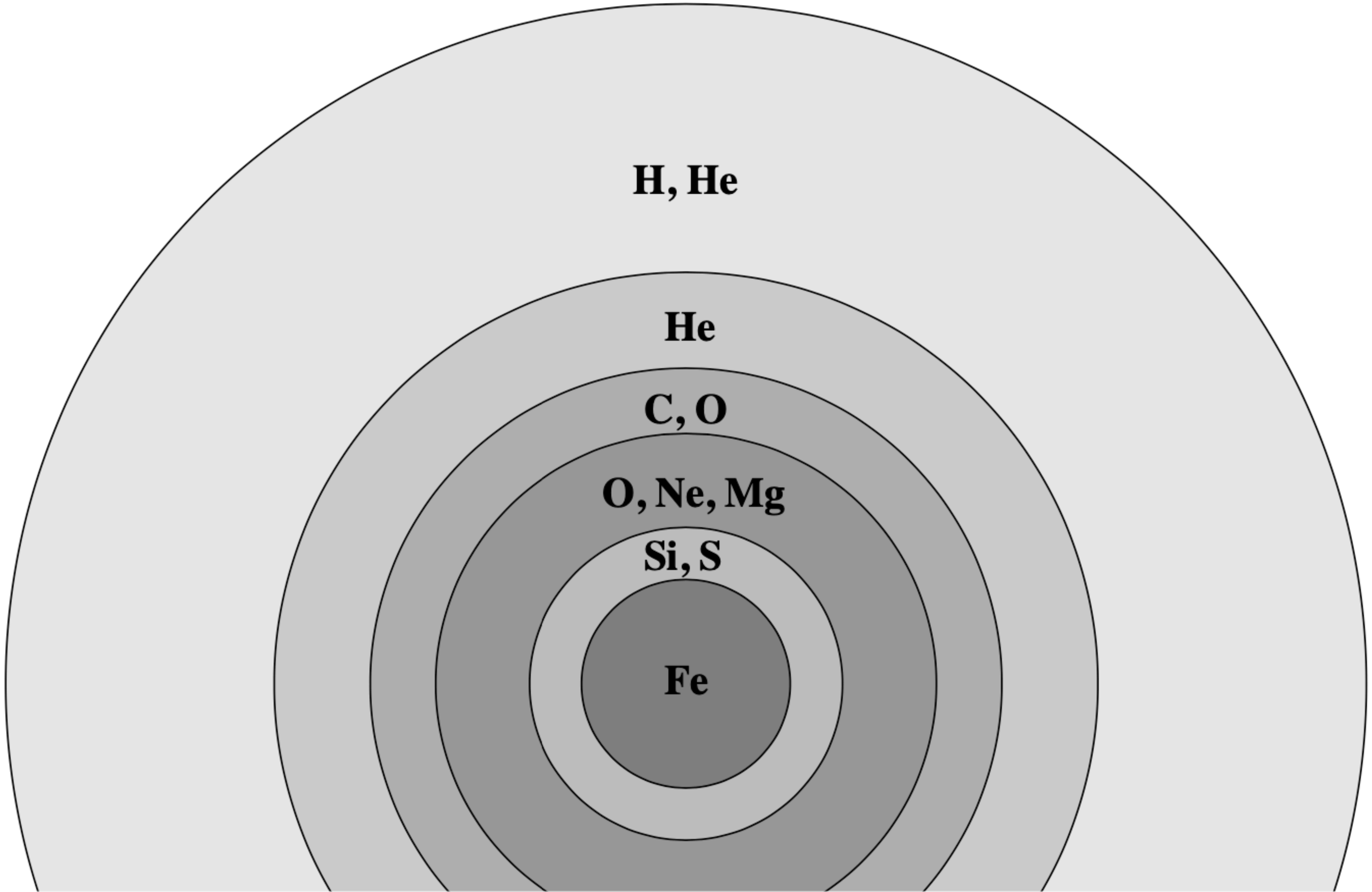
Chandrasekhar theory allows for relativistic electrons

$$M_{\mathrm{Ch}} = 1.459\left(\frac{2}{\mu_e}\right)^2 M_\odot,$$

Ideal case: fully degenerate electrons

Closer to 1.4 $M_\odot$ in nature

If stellar winds are not too strong, we end up with an onion-like structure and an Iron core



This final structure determines how the supernova explosion and/or compact object formation proceeds

Two core-collapse supernova classes:

Electron Capture

High densities lead to free e^-'s being captured through inverse β decay which reduces e^- degeneracy & pushes core out of hydrostatic equilibrium

Can occur in O-Ne cores with capture by ^{24}Mg and ^{20}Ne

Photo Disintegration

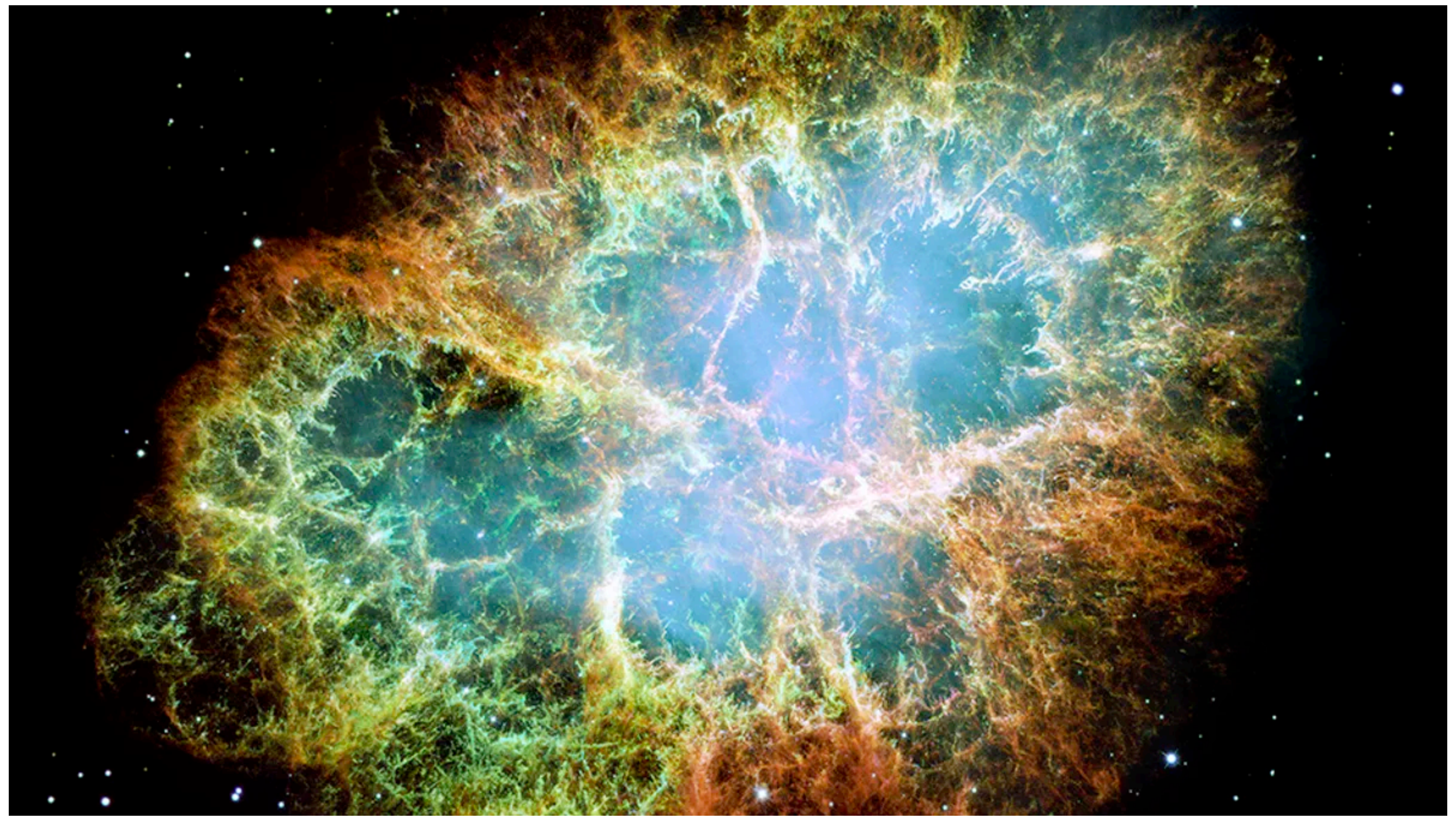
Fe core collapse produces $T \sim 10^{10}K$ such that photon energies become large enough to break ^{56}Fe into ^{4}He and neutrons.

This leads to a near free-fall collapse of the core

Two core-collapse supernova classes:

Electron Capture

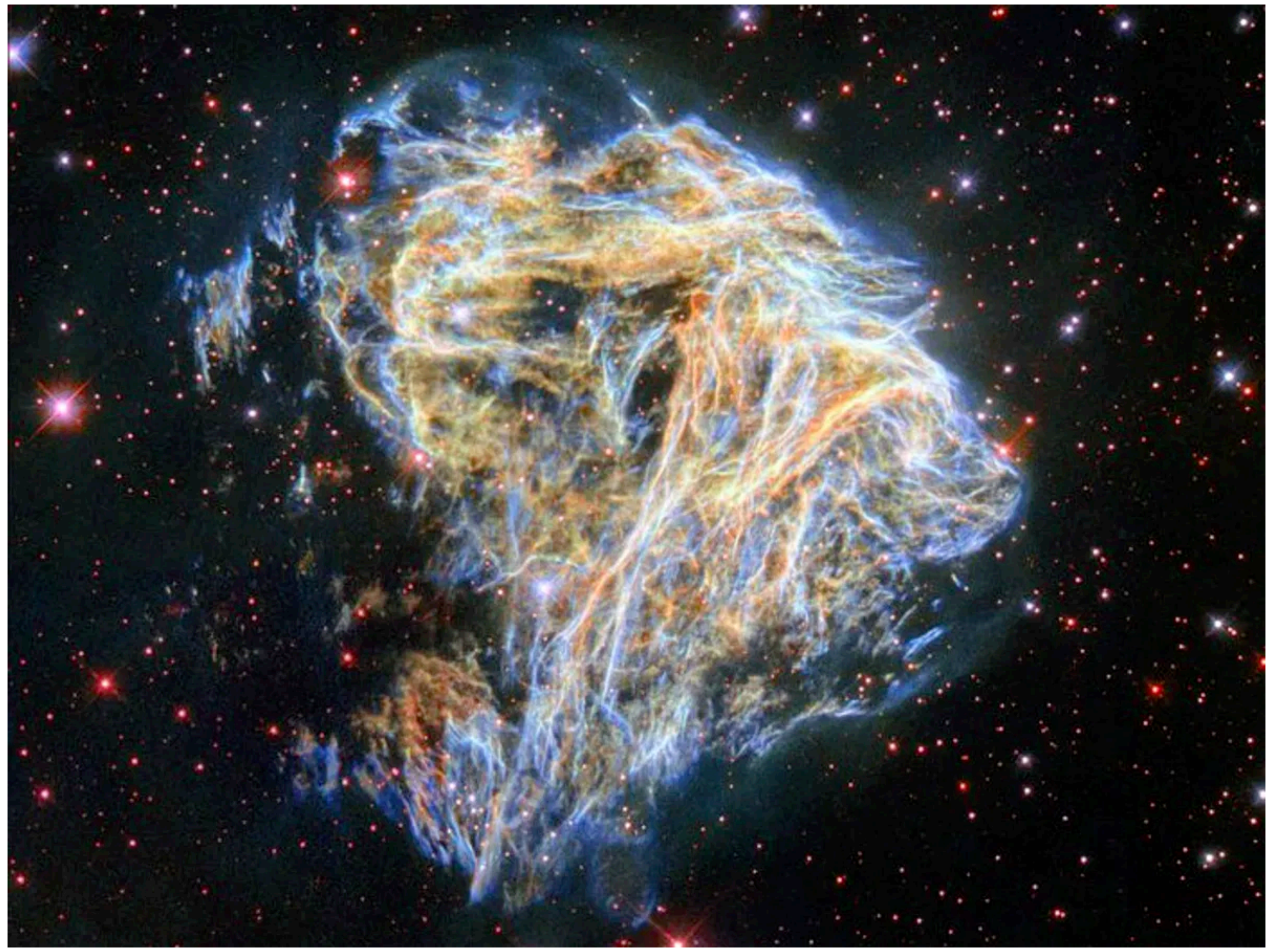
Largely spherically symmetric



SN 2018zd & the Crab
Hiramatsu+2021

Photo Disintegration

Thought to be asymmetric



DEM L 90

Energetics of core collapse:

$$E_{\rm gr} \approx -\frac{GM_{\rm c}^2}{R_{\rm c,i}} + \frac{GM_{\rm c}^2}{R_{\rm c,f}} \approx \frac{GM_{\rm c}^2}{R_{\rm c,f}} \approx 3 \times 10^{53}\,{\rm erg},$$

$R_{\rm c,i} \sim 3000\,{\rm km}$

$R_{\rm c,f} \sim 20\,{\rm km}$

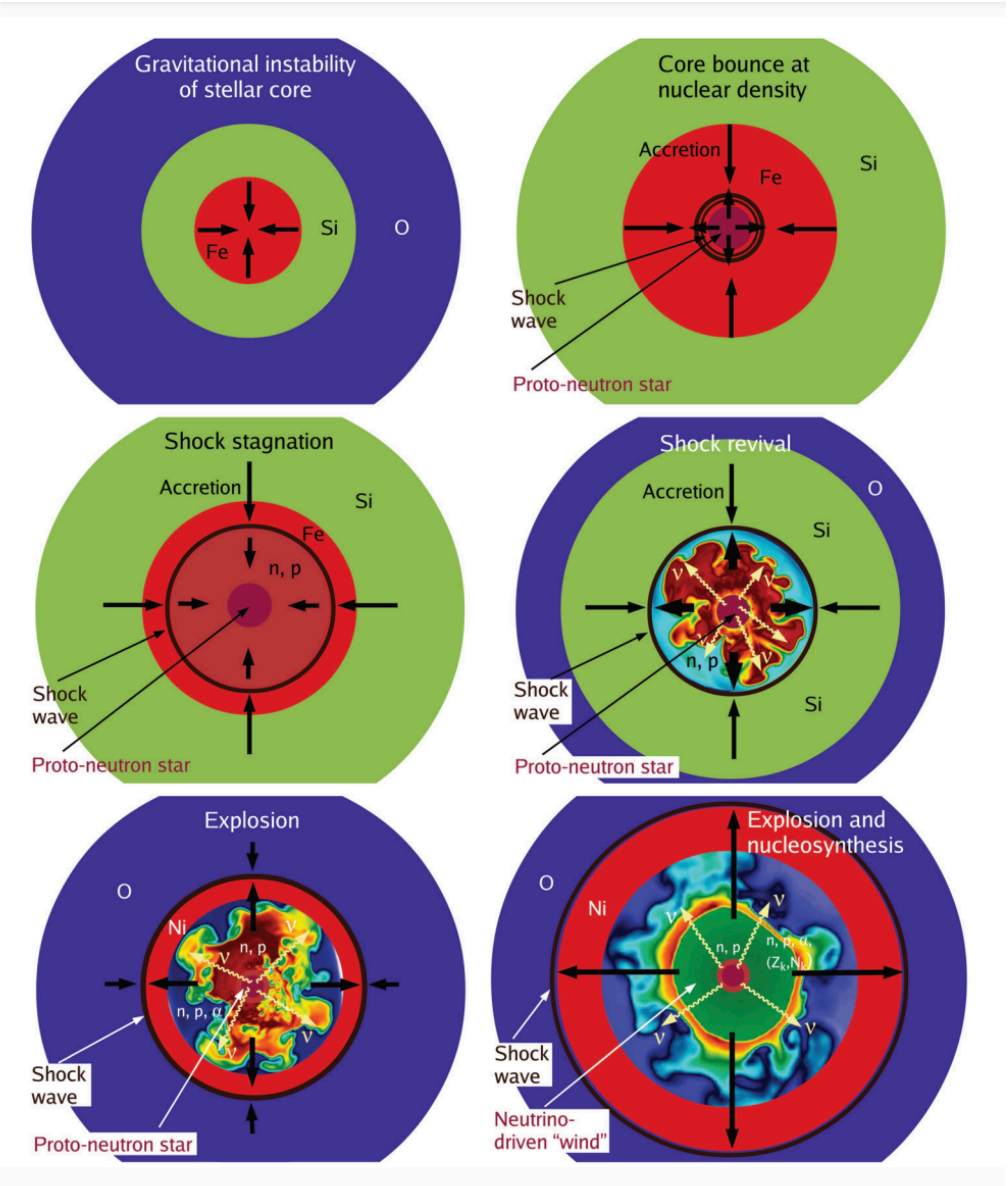
$\boxed{E_{gr} > E_{env}}$

$E_{kin,obs} \sim 10^{51}$ erg

$$E_{\rm env} = \int_{M_{\rm c}}^{M} \frac{Gm}{r}\,dm \ll \frac{GM^2}{R_{\rm c,i}}.$$

How do we convert E_{gr} to E_{kin}?

$E_{\rm env,realistic} \sim 10^{50}\,{\rm erg}$

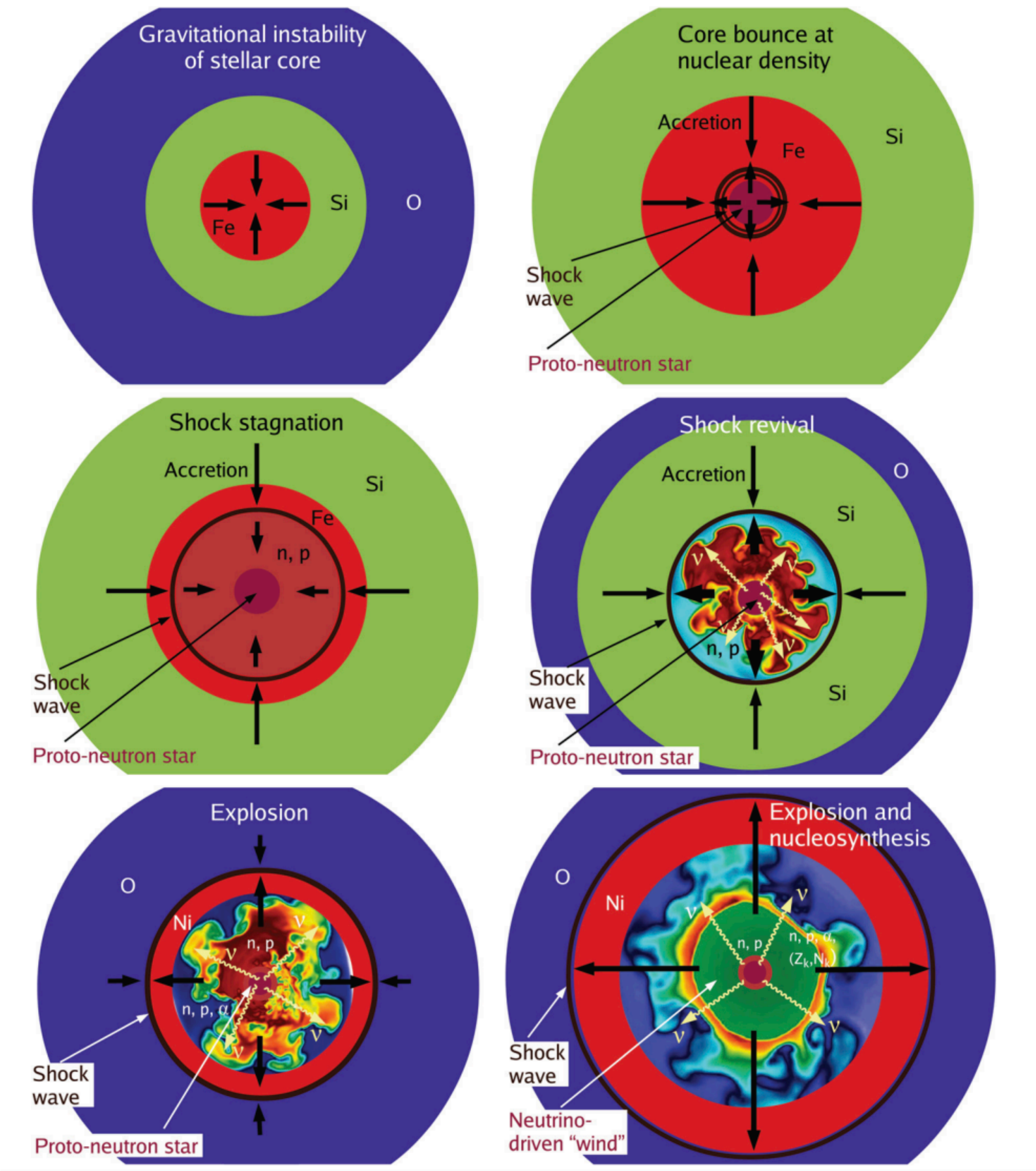


Core bounce vs supersonic infall leads to shock

As shock expands it photodisintegrates elements

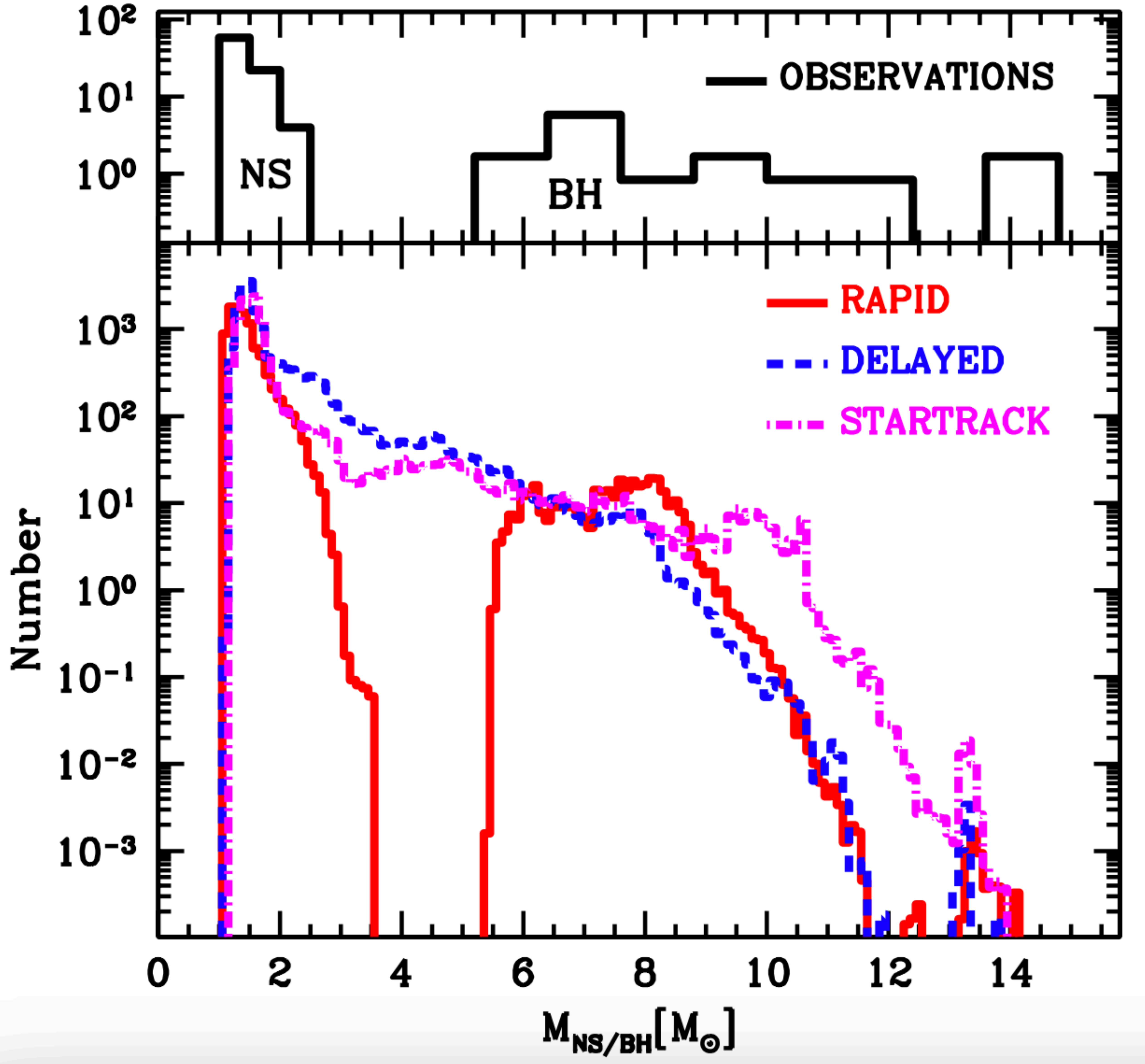
e^- captures lead to neutrinos which carry away energy

Shock stalls



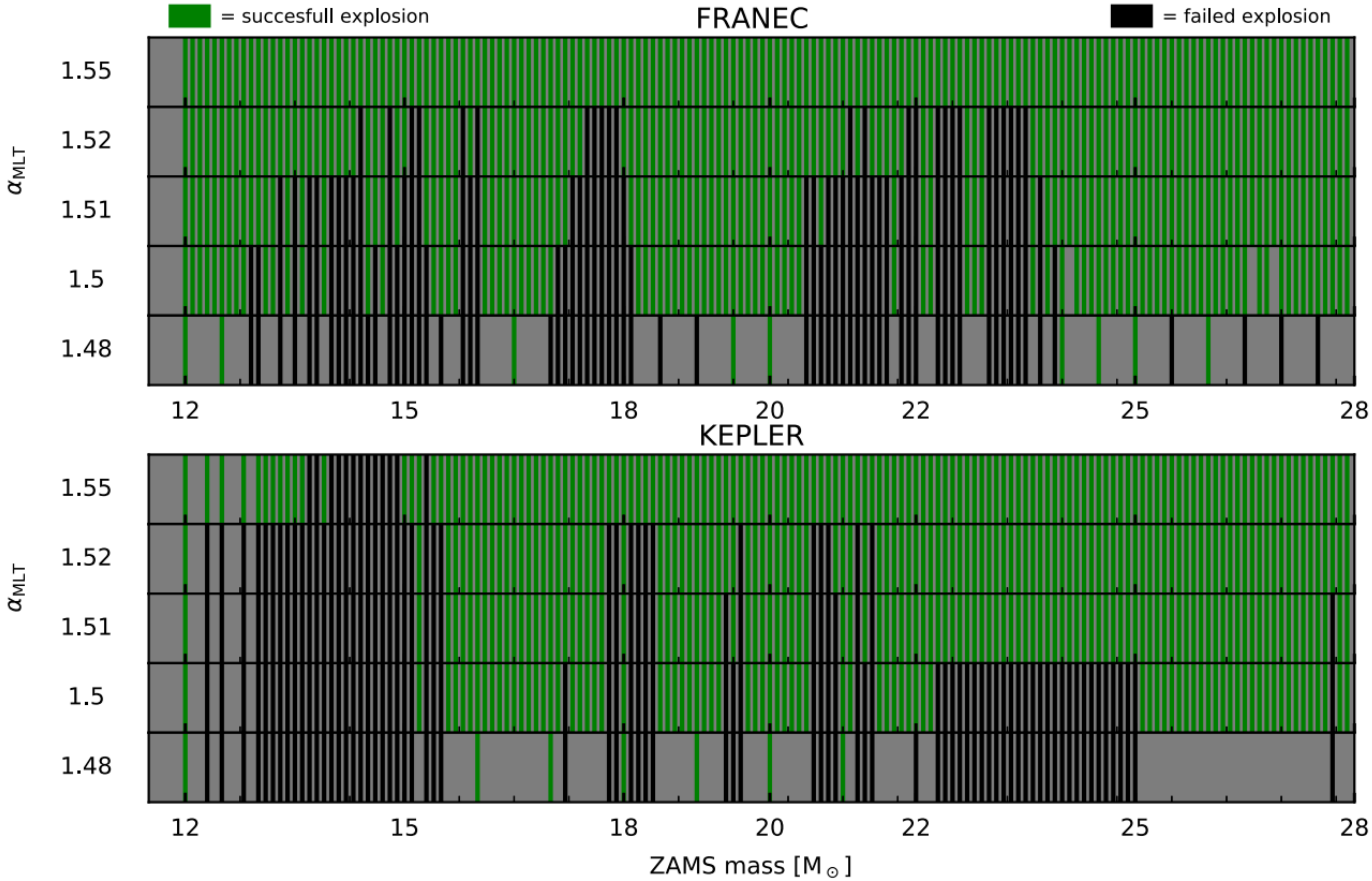
Infalling gas becomes opaque to neutrinos and neutrino production leads to convective instabilities which reignite the shock and eventual supernova

This is asymmetric and imparts a natal kick



If the shock is reignited rapidly — there is a gap between the masses of neutron stars and black holes

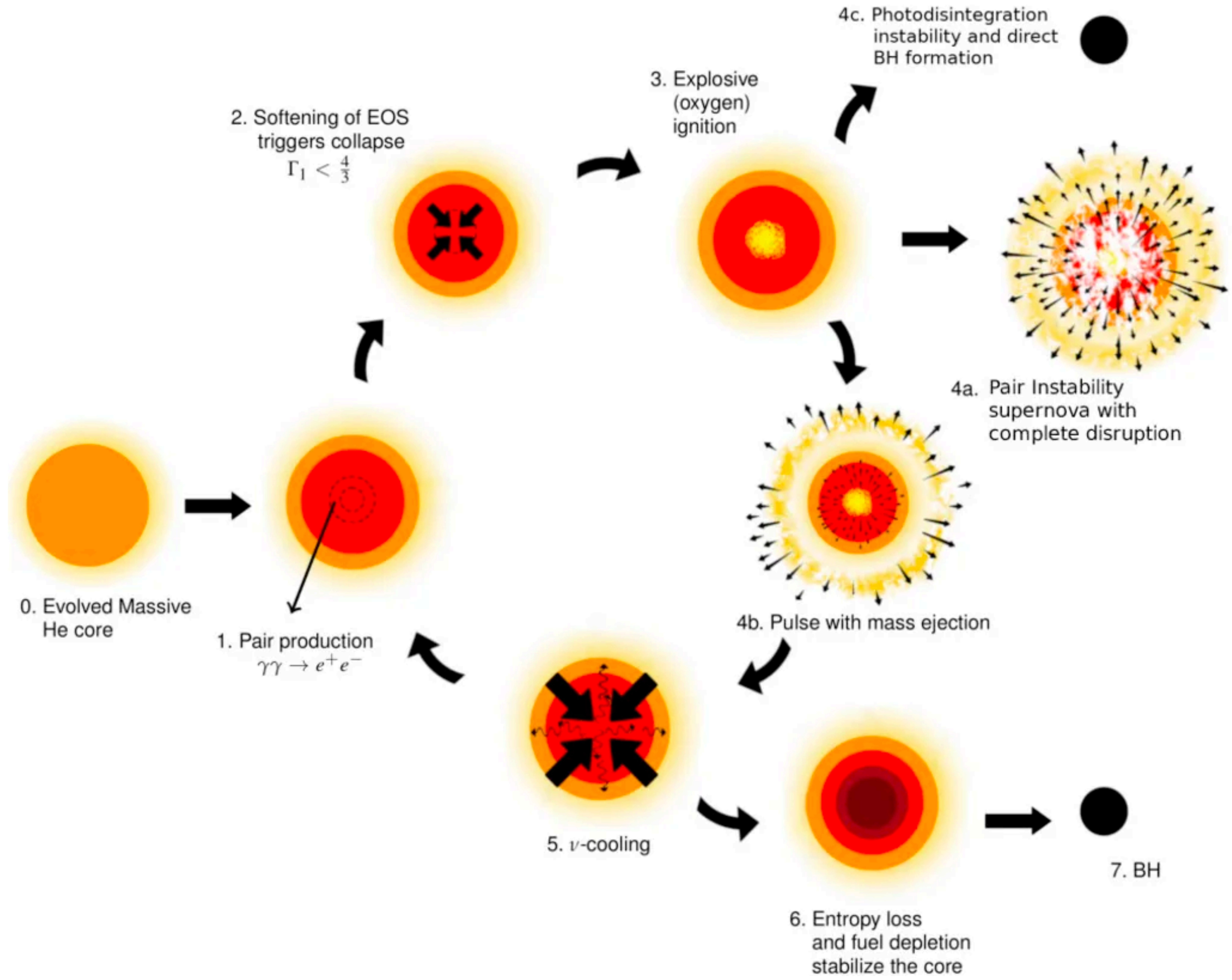
If it can happen on longer timescales the gap disappears



Explosion leaves behind neutron star

Failure leaves behind black hole

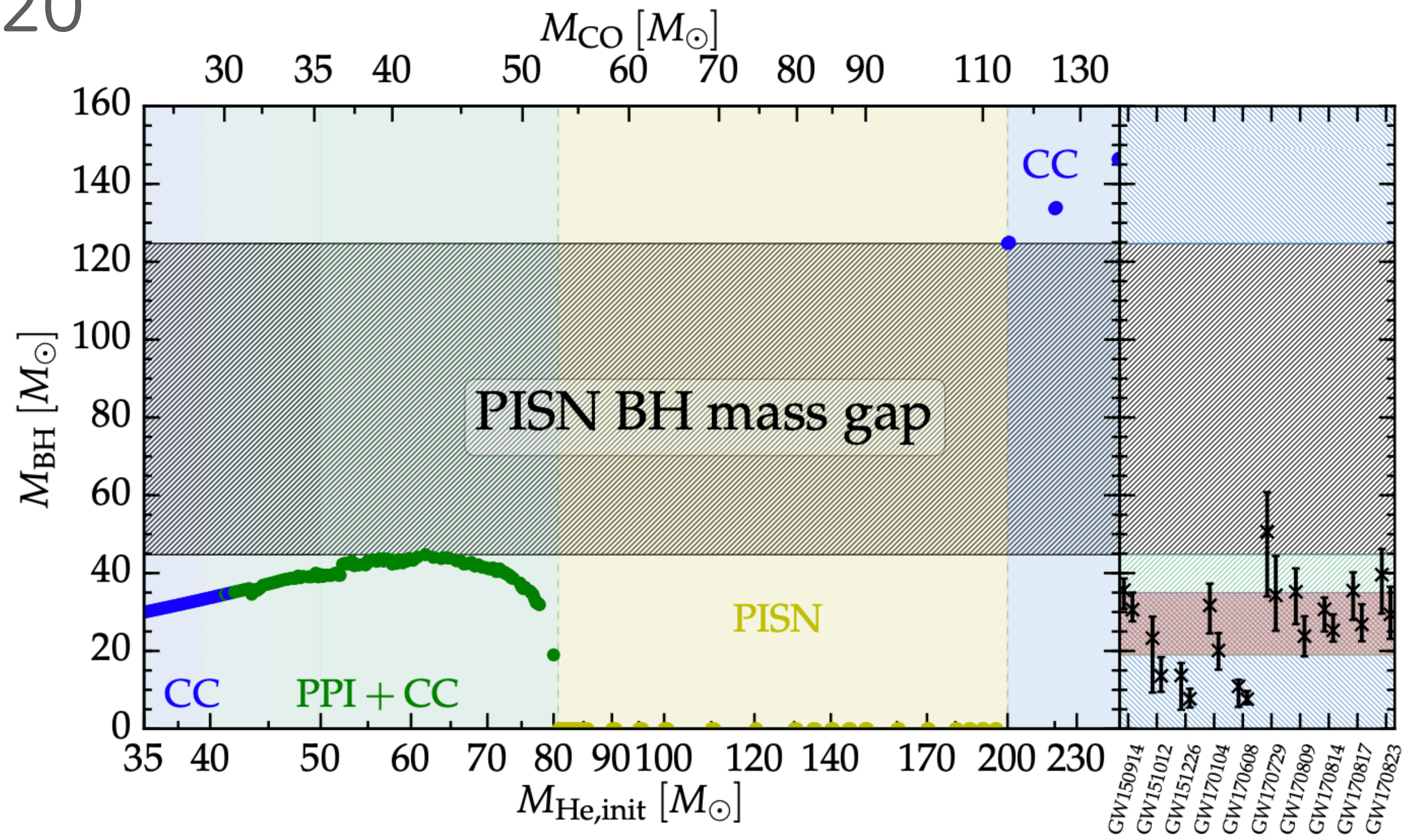
Boccioli+2023



Pair instability supernovae are special because they leave no remnant behind at all

Renzo+2020

Renzo+20



This creates an upper mass gap between 45-130 $M_\odot$

Project List 2

33-467: Astrophysics of the Stars and Galaxy

Fall 2023